

Geospatial Imputation of Urban Mobility Data with Self-Supervised Learning

Bin Han
University of Washington
bh193@uw.edu

Bill Howe
University of Washington
billhowe@uw.edu

Abstract

Missing values in urban data can be caused by sensor or software failures, data quality issues, interference from weather events, incomplete data collection, or varying data use regulations; any missing data can render the entire dataset unusable for downstream applications. In our work, we adapt image inpainting techniques to impute large, irregular missing regions in urban settings characterized by temporal dependency and spatial skew. To incorporate temporal information, we adapt computer vision techniques for image inpainting to operate on 3D histograms (2D space + 1D time) commonly used for data exchange in urban settings. To combat spatial skew of urban data — small dense regions surrounded by large sparse areas, we 1) train simultaneously in space and time, and 2) focus attention on dense regions by biasing the masks used for training to the dense regions in the data. We evaluate the core model and these two extensions using the NYC taxi data and the NYC bikeshare data, simulating different conditions for missing data. We show that the core model is effective qualitatively and quantitatively, that biased masking during training reduces error, and that the number of timesteps during learning exhibits a tradeoff between model performance and resolution of transient events.

Keywords: image inpainting, urban computing, spatial-temporal, missing data

1. Introduction

High-quality, longitudinal, and freely available urban data, coupled with advances in machine learning, improve our understanding and management of urban environments. Although conventional machine learning techniques are common in urban applications [17], neural architectures are opening new opportunities by adapting convolutional, recurrent, and transformer architectures to spatiotemporal settings [11]; see Grekousis 2020 for a recent survey [9]. For example,

spatio-temporal neural architectures have been used in predictions of rideshare demand [20], traffic conditions [16], and air quality [15]. But these models depend on access to complete, longitudinal datasets. Although the amount and quality of urban data available has dramatically increased with the proliferation of urban sensing and the public release of administrative data, missing data remains a challenge due to sensor or software failures [26], legal restrictions [1], or unusual events (adverse weather [6]). Missing values render datasets unusable, thus limiting the opportunity for understanding cities as the complex systems they are [10]. For example, [5] showed that missing data hinders the traffic prediction performance significantly. [4] demonstrated that without sufficient traffic flow data, the traffic control system is incapable of generating appropriate management strategies.

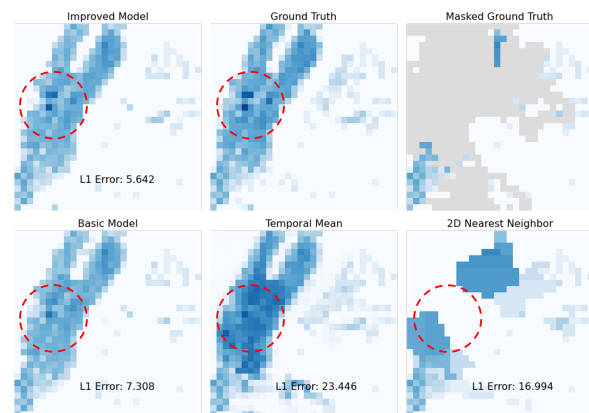


Figure 1: A histogram of taxi pickups in Manhattan. We reconstruct corrupted data in urban settings: Our model (upper left) uses biased masking and temporal context to capture local effects (red circle). The basic model (lower left) uses ordinary masking and is insensitive to local effects. Baseline methods that ignore space (lower middle) or time (lower right) are not competitive.

In this paper, we consider a neural data imputation strategy based on masking out corrupted regions and using a trained model to reconstruct the masked region.

These masks are necessarily large, irregular, and extend in both time and space. By modeling arbitrary missing masks, we afford user control: any areas can be masked and reconstructed. Additionally, the proposed strategy represents a general self-supervision framework for urban regions and is applicable to any dataset of human activity governed by population density.

Spatiotemporal interpolation of missing data has been widely studied in the earth sciences, especially in remote sensing where weather effects can obscure measurement [26]. Conventional statistical approaches to impute missing values, such as global/local mean imputation, interpolation, and kriging, are essentially linear, and therefore have limited ability to capture the nonlinear dynamics needed to impute large irregular missing regions. Neural image inpainting techniques can recover missing patches via training on large datasets of independent images, such that the reconstructed images appear smooth and realistic.

These approaches have shown promising results with global climate data [23], but have not been adapted to the urban setting in which data are not smooth functions of space and time, but are rather histograms of events constrained by the built environment. This data property leads to the challenge for imputation in the urban setting, which is *skew*: urban data tends to follow population density patterns — small dense regions surrounded by large sparse areas. These population patterns can dominate the learning process and fool the model into ignoring numerical accuracy in dense regions, even while aggregate error may remain low. Therefore, the overall research question we aim to answer in this work is: Can the spatially skewed, time-varying nature of urban mobility data be exploited to improve over existing methods for missing urban data imputation?

Following the literature, we represent spatiotemporal event data in a 2D or 3D raster form (e.g., a histogram). Our basic model uses the partial convolution approach from Liu et al [13] to handle the irregular boundaries of missing data (e.g., districts), which focuses model attention on the valid regions while shrinking the masked region, layer-by-layer, to obtain a complete prediction. We extend the basic model to the 3D spatiotemporal setting to capture the temporal information. To combat skew, we 1) bias the training process to focus on populated regions by seeding the mask in non-zero areas; 2) use 3D convolutions and vary the number of timesteps in each 3D training sample to capture transient events. Together, these two techniques complement each other: biased masking focuses attention on dense regions, and 3D convolutions with a large chunk size focus attention on sparse regions.

We evaluate these techniques on the NYC taxi

data (a popular dataset for its coverage and quality) and a NYC bikeshare dataset (less dominated by the built environment). We find that the basic model is effective for urban data imputation, while biased masking reliably reduces error over random masking, both globally and locally. Additionally, we find that the number of timesteps per training sample exhibits a tradeoff: too few timesteps and the model ignores transient patterns, while too many timesteps significantly increases training time without enhancing the inpainting results. We evaluate specific local scenarios (high-traffic locations, low-traffic locations, high-variability locations, anomalous events) to reflect real-world use cases. Besides direct imputation, additional applications may include forecasting, quality assessment, transfer learning between cities, and uncovering relationships between variables. In summary, we make the following contributions:

- We evaluate a basic model adapting image inpainting techniques to urban histograms characterized by skew and sparsity effects due to constraints by the built environment, demonstrating qualitative and quantitative accuracy relative to classical methods.
- We improve on this basic model by extending to the 3D spatiotemporal setting to better recognize transient events; we analyze the training time and performance tradeoffs of varying the number of timesteps per training sample.
- We propose a self-supervised training process called biased masking to encourage the model to attend to dense population regions and thereby improve accuracy on the highly dynamic regions typical in urban environments; we show that biased masking reliably improves convergence.
- We evaluate these techniques on two real mobility datasets (NYC taxi trips and NYC bikeshare trips), both globally and locally in varying traffic conditions, weather events, and disruptions. Finally, we show that the model can be used to remove or synthesize anomalous events through targeted masking.

2. Related Work

Image Inpainting Image inpainting involves synthesizing missing pixels in images, such that the reconstructed images are visually credible and semantically realistic. In computer vision, there are two broad categories of inpainting techniques. The first category contains diffusion-based or patch-based methods, which utilize low-level image features to recover the missing pixels. We focus on the second category, which contains learning-based methods that generally involve the training of deep neural networks.

Learning-based methods are trained to learn image patterns from large datasets of unrelated images to recover missing regions and preserve semantics. Pathak et al.[18] proposed the context encoder, combining a CNN with a generative adversarial network in an encoder-decoder architecture trained using both ℓ_2 reconstruction loss and generative adversarial loss in the objective function. Yu et al. [25] proposed a two-stage coarse-to-fine model architecture and incorporated contextual attention layer to attend to related features from spatially distant regions. Liu et al. [13] proposed partial convolution, allowing inpainting models to be used on irregular holes rather than just rectangular missing regions. On top the work of partial convolution, Zhou et al.[27] incorporated dual spatial attention modules into the U-Net architecture, which can capture the correlations between facial textures at different scales. We refer readers to the following survey paper [14] for details of more inpainting methods, both diffusion- and learning-based.

This recent trajectory in image inpainting involves reducing or eliminating perceptual artifacts such as discontinuous edges and blurred patches using new loss terms, image preprocessing, or training regimes that favor subjective quality over numerical accuracy. For example, Xu et al. [24] both propose extensions to partial convolutions to repair blurred boundaries between missing and valid regions. Since our focus is on numerical accuracy and downstream utility of the synthesized data rather than perceptual smoothness, and because we aim to study model-agnostic extensions related to the urban setting, we base our approach on the work from Liu et al. [13].

Geospatial Missing Data Imputation Classical spatio-temporal interpolation methods, generally variants of inverse-distance or nearest-neighbor weighting [7], kriging [2], or matrix factorization [8] are variations of linear methods that do not attempt to (and cannot) interpolate within large, arbitrary, irregular regions, and typically do not seamlessly consider both space and time. Gong et al. [8] solve multi-variable non-negative matrix factorization to impute urban data, but assume the availability of multiple variables and do not consider arbitrary irregularities. Zhang et al. [26] proposed unified spatial-temporal-spectral deep CNN architecture to recover the missing regions in satellite images that were caused by atmospheric conditions. Kang et al. [12] restored the missing patterns of sea surface temperature (SST) from satellite images. These methods focus on remote sensing data, which represents a continuous field rather than event-based data constrained to the built environment.

3. Methods

We articulate our imputation methods in this section, including mobility datasets (3.1), model architecture (3.2), loss function (3.3), biased masking technique (3.4), and training details (3.5).

3.1. Representative Datasets

We consider two mobility datasets: NYC taxi data and NYC bikeshare data. Although potential *applications* of the proposed model are widely available, datasets on which to *evaluate* the model are rare: we need longitudinal coverage to provide ground truth, sufficient complexity to study both global and local fidelity, and public availability for reproducibility. For example, census data tends to be too coarse and lacks temporal resolution. The two mobility datasets satisfy all of these requirements, and are representative of human mobility in any urban region. The datasets are:

- **NYC Taxi Data.** NYC taxi trip data were collected from NYC Open Data portal from 2011 to 2016¹. The year 2011 — 2015 cover the trips throughout the entire year, while 2016 only covers the first half of the year until June 30. The raw data are presented in tabular format. Each record from the data summarizes the information for one single taxi trip, which contains the longitude and latitude of the location where the taxi took off. Each record can be viewed as one taxi demand count.
- **NYC Bikeshare Data:** NYC bikeshare data were collected from NYC DOT from 2019 to 2021 portal.² All three datasets cover the bike trips throughout the entire year. Similar to the taxi data, the raw data are presented in tabular format. Each data point summarizes the information for one single bike trip, including the longitude and latitude of the location where the bike was unlocked. Each record can be viewed as one bike demand count.

We aggregate both datasets into a 3D histogram by defining a rectangular region, then binning mobility events into a regular grid to create a 2D histogram amenable to image techniques. These 2D images are stacked to create 3D blocks. The temporal depth of the block is a parameter we study in this paper. We defined the NYC region as shown in Figure 3. At this resolution, the region has a relatively balanced coverage of areas with different levels of demand counts: high demand in Manhattan and near airports, and low demand elsewhere. This skew is common in urban applications and represents both an opportunity and a challenge for

¹<https://opendata.cityofnewyork.us/data/>

²<https://ride.citibikenyc.com/system-data>

neural prediction: the patterns are relatively stable, but the sparse regions can dilute the learning process.

We then define a 64×64 grid over the region of interest. We chose this resolution to be fine enough to capture activity at the city block scale (each pixel is only 2-3 city blocks) while minimizing training time. We did not study how varying the resolution affects results, as previous works in computer vision have shown that comparative model performance is relatively insensitive to resolution [22]. Then, for each hour of each day, we count the number of taxi/bike trips that began within each pixel. The value is commonly interpreted as an estimate of demand. We do not consider multiple resolutions in this paper. After processing, we have 30,648 training images and 3,620 test images in NYC taxi data, and we have 25,560 training images and 720 test images in NYC bikeshare data.

3.2. Inpainting Model

In this section, we describe the basic model for using partial convolutions for inpainting spatiotemporal urban histograms. Each input sample consists of a *masked* region with unknown, corrupted, or inaccurate values to be reconstructed and a *valid* region with known values. The task is to predict values in the masked region to match the original image. Training is self-supervised by creating random masks for any input image. We adapt the architecture from Liu et al.[13], which proposed partial convolutional layers to accommodate irregular masks. Partial convolutions ignore the masked region, but the mask “shrinks” after each partial convolution layer by unmasking pixels near the boundary. After several layers, the all values are unmasked and the entire output is considered valid. We use the U-Net architecture with skip connections [19], and replace convolution layers with partial convolution layers.

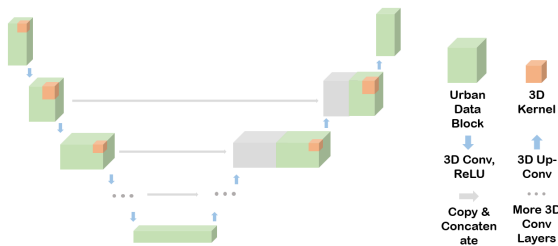


Figure 2: The model architecture is a U-Net extended to 3D, partial convolutional layers [13] to ignore masked regions during training. In the decoding branch, multiple 3D up-convolutional layers are utilized and skip-connections are applied. In total, there are six encoding layers and six decoding layers.

Web images only contain 2D information (ignoring RGB channels), but urban histograms vary in both space

and time. As a result, our training data is essentially one massive 3D block rather than a large number of independent training images. We therefore have a design choice of how to “shred” this block into training samples. In this paper, we consider only the temporal extent in 3D; varying spatial resolution, bounds, or overlap during rasterization of the source data is left for future work. We therefore extend all convolutional layers, inputs, and masks, to 3D. The inputs are 3D image blocks of dimension $T \times W \times H$, where T represents the temporal dimension. The masks are also in 3D blocks with the same shape as the image block. The architecture is shown in Figure 2.

3.3. Loss function

We use ℓ_1 loss as the objective function for pixel-wise reconstruction accuracy. It bridges the absolute gap between the reconstructed value and the ground truth. We adopt the following notation:

- $\mathbf{I}_{gt} \in R^{T \times W \times H}$: the block of ground truth images. T represents the temporal dimension of the block.
- $\mathbf{I}_{out} \in R^{T \times W \times H}$: the block of reconstructed images.
- $\mathbf{M} \in R^{T \times W \times H}$: the block of binary masks.
- $N_I = T * W * H$: number of pixels in the image block.
- N_{valid} : number of valid pixels in the image block.
- N_{hole} : number of missing pixels in the image block.

We separate the loss function for valid and hole regions. Even though the valid region has available data and we typically would not use the predicted values in practice, we include this loss during training to improve continuity across mask boundaries (therefore improve overall error). The ℓ_1 loss is calculated as: $\mathcal{L}_{total} = \mathcal{L}_{valid} + \lambda \mathcal{L}_{hole}$, where $\mathcal{L}_{hole} = \frac{1}{N_{hole}} \| (1 - \mathbf{M}) \odot (\mathbf{I}_{out} - \mathbf{I}_{gt}) \|_1$ and $\mathcal{L}_{valid} = \frac{1}{N_{valid}} \| \mathbf{M} \odot (\mathbf{I}_{out} - \mathbf{I}_{gt}) \|_1$.

3.4. Biased Masking

By default, masks can be generated by randomly select a starting point in the image and then conducting a random walk for a fixed number of step. We call this process **random masking**. However, since urban data is constrained by the built environment and is therefore highly skewed toward populated areas, random masks tend to include a large number of zero-valued cells, squandering opportunities to learn from the steep gradients in dense, high-traffic regions; To focus attention on populated areas, we use a **biased masking** approach: 1) Given an input image, apply Gaussian blur to blend the pixel values and increase the region of potential starting points. 2) Select a threshold (e.g., 90 percentile of the image values) to identify populous regions. In our experiments, the 90

percentile was set by considering multiple values on 24 representative timesteps via manual inspection. Setting the threshold to the 90th percentile preserved important high-traffic features such as the airport while excluding low-traffic areas in the suburbs. This threshold can be set by a domain expert following a similar heuristic procedure, or set via hyperparameter tuning techniques. 3) Randomly select a starting location from one of the detected areas and generate masks via random walk. The probability of selecting one of the detected areas is proportional to the size of the area. The biased masking approach makes the learning problem more challenging by increasing “contrast”: ensuring that masks tend to include dense, dynamic regions, but also include sparse, stable regions. To compare the performance of the two masking approaches, we generated two masks (one random and one biased) for each training sample.

3.5. Training

With NYC taxi data, we trained the models on both mask types — random and biased, and with different temporal dimension $T = \{1, 2, 3, 5, 7, 10, 15\}$. Based on initial experiments on both mask types and at lower temporal chunk sizes, we found that $\lambda = 12$ offered effective performance; we fix λ to be 12 for all experiments on the taxi data. With NYC bikeshare data, we trained models with both mask types, $T = \{1, 2, 3\}$, and $\lambda = 4$. The batch size and initial learning rate are set to 16 and 0.01 respectively. Learning rate decays every 500 training iterations at rate of 0.9.

4. Experimental Evaluation

We consider the following sub-questions to evaluate the model capabilities:

- (Q1) Is the core 3D model qualitatively & quantitatively effective at inpainting missing data? (Section 5.1, Figure 3, Table 1)
- (Q2) Does increasing the number of timesteps per training sample generally improve performance? (Section 5.2, Figure 4)
- (Q3) Does the model faithfully reconstruct local, dynamic conditions in specific areas of interest? (Section 5.3, Figure 6)

Q1 helps us understand the effect of biased masking on overall performances, while Q2 investigates the effects of incorporating temporal information. Q3 helps demonstrate the utility of models in real-world scenarios. Unless otherwise stated, we evaluate the model on the test set using $\ell_{1, \text{hole}}$, which is the sum of the absolute value of the difference between the ground truth and predictions at the masked positions only. We compare our models with baseline statistical methods:

- **Temporal Global Mean:** On the training data, we calculate the average taxi demand at each pixel, for each hour of the day. On the test data, we assign each masked pixel the corresponding global mean computed from the training data.
- **Nearest Neighbor (NN) Interpolation:** We assign each masked pixel the value of the nearest unmasked pixel. We experimented with both 2D and 3D implementations using scipy.³
- **RBF Interpolation** We interpolate using radial basis functions (RBF) on observations at points sampled outside the masked region. We experimented with both 2D and 3D RBF interpolation with RBF Python implementation.⁴

We considered 3D kriging, but found the poor scalability to be prohibitive: the estimated time to complete the computation for an experiment with $T=2$ was about two weeks on a typical platform! Moreover, kriging is a local linear method, and we have no reason to believe that it can reconstruct data across large, irregular regions. Another approach, which we did not study, is to use physics-based models based on computational fluid dynamics [3] or agent-based models that directly encode human behavior [21] to capture macro traffic dynamics. These approaches can “fill” large missing regions, but must be designed separately for each application.

5. Results

5.1. Model Effectiveness (Q1)

We find that for both taxi and bikeshare datasets the proposed model faithfully captures qualitative visual patterns and also significantly outperforms baseline methods on multiple metrics.

5.1.1. Qualitative Analysis We first present some visual examples of inpainting results on NYC taxi data in Figure 3. The left figure shows taxi demand at two different hours of the day (2PM and 2AM). From left to right, we show the ground truth, the (biased) mask, the mask applied to the ground truth, and the reconstructed image. The inpainting model was trained with 5 timesteps per training sample and with biased masking. For all hours and all masks, the model is effective at reconstructing missing data, even when the majority of the signal is obscured. The reason is clear: the patterns are sufficiently stable from timestep to timestep as to allow the model to infer missing values from temporal patterns as well as spatial patterns. The

³<https://docs.scipy.org/doc/scipy/reference/generated/scipy.interpolate.griddata.html#scipy.interpolate.griddata>

⁴<https://github.com/treverhines/RBF>

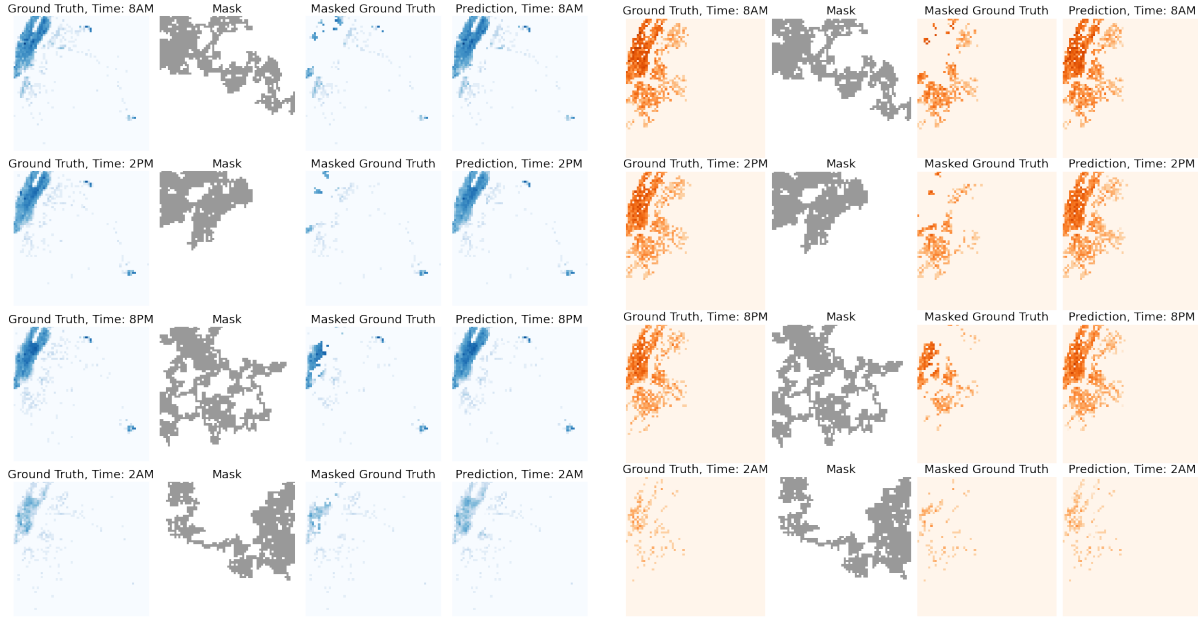


Figure 3: Reconstructed results of taxi demand images (Left) and bike demand images (Right) at different hours time trained with biased masking and 3D partial convolutions ($T=5$ for taxi data and $T=3$ for bikeshare). From left to right, each column displays the ground truth image, mask, masked ground truth, and reconstructed data. From top to bottom, each row presents the taxi demand at 2PM and 2AM, respectively.

model is also responsive to the time of day: We see fewer rides at 2AM than at 2PM, as expected, suggesting that the model has learned temporally local patterns as opposed to relying on global spatial patterns. The transition across the mask boundary is also smooth, suggesting the model was able to consider local spatial patterns appropriately. Overall, we find that the model is perceptually effective at reconstructing missing values, even in challenging cases. The right plot in Figure 3 shows corresponding results for bikeshare data. We observe similar observations as the results from taxi data — at all times of day and for all masks, the reconstructed images are visually similar to the ground truth images, indicating the consistent effectiveness of our model.

5.1.2. Quantitative Analysis Table 1 contains quantitative results of baseline models and our neural models in different evaluation metrics. We observe that: 1) Our neural models, trained with either masking type or with any temporal dimension, always outperform the baseline models. The 2D baseline models that ignore the temporal dimension are especially ineffective. Global mean ignores spatial effects and just models a function $pixel, hour \rightarrow value$. 2D- and 3D- nearest neighbor methods perform poorly when the nearest neighbors may be far away; 2D- and 3D-RBF methods assume relatively uniform sampling across the region, which is not possible in our setting of wide-area missing data. 2) At $T=5$ and 7, our method performs similarly and

achieves the best performances — almost 50% lower ℓ_1 error and 66% lower ℓ_2 error than the best baseline. 3) SSIM does not significantly distinguish different models; while popular in image inpainting, this metric is designed to capture perceptual similarity of natural images, which are not relevant for the spatiotemporal aggregations we study. 4) The model training time increases by about 9 minutes for every additional hour included in a chunk. At $T=5$, the model takes 55

Model	Mask Type	$\ell_{1,hole}$	$\ell_{2,hole}$	SSIM	PSNR	Time (m)
G. Mean	-	1.2644	55.3298	0.9973	61.4880	<5
2D-RBF	-	3.1442	284.8807	0.9890	54.8346	70
2D-NN	-	3.1179	318.6575	0.9884	54.0717	<5
3D-RBF	-	1.6653	94.7708	0.9956	57.9921	>24h
3D-NN	-	1.3632	84.0529	0.9964	59.1652	<5
3D-RBF	-	1.6653	94.7708	0.9956	57.9921	>24h
Ours						
$T = 1$	biased	0.9081	37.8468	0.9984	62.4268	18
	random	0.9406	40.3730	0.9983	62.4679	18
$T = 2$	biased	0.8551	32.6429	0.9986	63.1815	27
	random	0.8979	35.2923	0.9985	63.1056	27
$T = 3$	biased	0.7847	25.8374	0.9987	63.6445	35
	random	0.7950	26.4765	0.9989	63.7221	35
$T = 5$	biased	0.7196	18.7080	0.9991	64.4028	55
	random	0.7606	20.6116	0.9990	64.1000	55
$T = 7$	biased	0.7185	18.6746	0.9990	64.3407	75
	random	0.7489	20.0100	0.9990	64.2656	75
$T = 10$	biased	0.7537	24.8383	0.9986	63.3329	75
	random	0.7820	26.1138	0.9985	63.1288	75
$T = 15$	biased	0.7729	25.3386	0.9985	63.1885	140
	random	0.7849	21.9446	0.9989	63.8721	140

Table 1: Model training time and performance.

minutes to train. The baseline heuristic-based methods — global mean and 2D- and 3D-NN — are very fast but very inaccurate given that they do not attempt to model global dynamics. The 3D-RBF method is inefficient: T=2 required over 24 hours to train.

5.2. Temporal Dimension Tradeoff (Q2)

Figure 4 shows the prediction errors for NYC taxi data, evaluated on random masks (top plot) and biased masks (bottom plot). The y-axis is the ℓ_1 loss considered for the masked region only (“Hole”). The x-axis varies the number of timesteps included per training sample (Temporal dimension), ranging from 1 to 15. (a) When tested with random masks, the average mask covers the entire region, concentrated at the center. Models trained with biased masking reduces error at all sizes. The ℓ_1 error decreases as the number of timesteps increases up until T=7, then starts to increase again (T=5 and T=7 have similar performances when trained with biased masking.) At T=2, the model begins to make use of the temporal dependency between the data by applying 3D convolutions. With both biased and random masking, the ℓ_1 loss decreases sharply when T changes from 1 to 5. (b) When tested with biased masks, the average masked cells are concentrated at the upper left due to the bias toward populated regions. The plot has a similar U-shape as that of random masking.

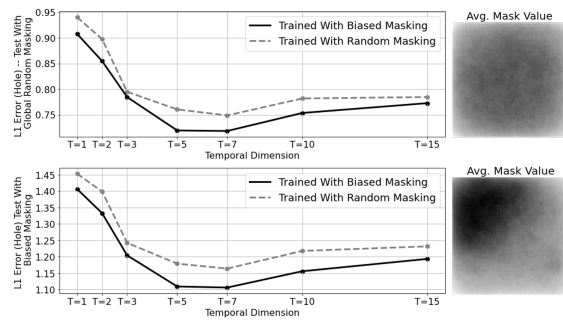


Figure 4: Evaluation of models trained with biased masking against those trained with random masking, at seven temporal dimensions, with two different masking scenarios — random and biased masking.

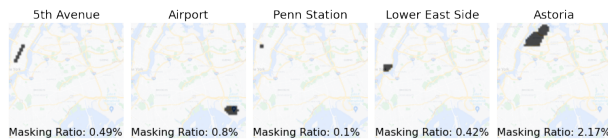


Figure 5: Scenario masks overlaid on NYC map. Annotation: The ratio of masked-to-unmasked area.

5.3. Scenario-Based Evaluation (Q4)

Spatiotemporal patterns of missing data in practice are unlikely to resemble random walks. Instead, outages will correlate with environmental features: sensors may fail in certain weather conditions, transient events may prevent data acquisition, or legal restrictions on data availability may follow political boundaries. To demonstrate applicability in real-world scenarios, we evaluate on specific locations under varying conditions. We tested five different scenarios to cover varying spatial locations, temporal variances, and social events. The five scenarios include masking 5th Avenue, Penn Station, airport, lower east side, and Astoria (Figure 5).

5th Avenue and Penn station are representative of busy and dense areas with heavy traffic. 5th Avenue can also show the impacts of certain social events on traffic patterns: The Pride Parade showed an anomalous intervention where traffic was zero on the parade route. Lower East Side is away from central Manhattan, with relatively lighter traffic than the first two cases. The scenario of airport and Astoria represent sparse regions with light traffic. We chose two periods in 2016 to cover temporal variance: 2/1 to 2/15 (capturing a snowstorm that dramatically affected traffic city-wide; see Figure 6) and 6/18 to 6/25 (capturing the Pride Parade near 5th avenue that affected traffic in a more localized area.)

Scenarios	G.T.- Biased	G.T. - Random	G.T. - Mean
02/01/2016 — 02/15/2016			
5th Avenue	4.2	6.2	17.0
Penn Station	19.3	33.5	30.0
Lower East Side	2.5	2.8	8.2
Airport	2.3	1.6	1.8
Astoria	0.8	0.7	0.4
06/18/2016 — 06/30/2016			
5th Avenue	3.6	4.8	22.53
Penn Station	21.6	37.5	30.0
Lower East Side	1.7	2.1	7.4
Airport	2.4	1.9	2.0
Astoria	0.8	0.7	0.4

Table 2: Average absolute error between the predicted values and ground truth, over the missing regions, and during the selected evaluation periods.

We test three inpainting models: our model trained with biased masking at T=5, the same model but trained with random masking at T=5, and the global mean approach. We plotted the ground truth and predicted values at the average pixel level in the missing region, for each hour during the selected periods. The visualizations are provided in Figure 6. The average absolute errors between the ground truth and predicted values, over the missing region and during the evaluation periods, are reported in Table 2. We have the following observations:

- For the traffic-dense location (5th Avenue, Penn

Station, and Lower East Side), our models, regardless of masking strategy, match ground truth better than the temporal mean, indicating that the model is learning localized effects rather than just a longitudinal average. This benefit holds for both evaluated periods, as shown in both Table 2 and Figure 6. For the airport and Astoria scenarios, the temporal mean is very slightly better.

- Table 2 shows that for both evaluation periods the model trained with biased masking has lower average error than the model trained with random masking, except in the scenario of the airport in June. Since random masking favors sparse areas, this difference is expected.
- During the snow days (02/05-02/08/2016), it is expected that the traffic in the dense regions would be significantly impacted, which can be supported by the trough seen from the ground truth line in the scenario of Penn Station (other scenarios are not heavily impacted by the snow.) The model trained with biased masking is responsive to the irregular traffic caused by extreme weather, unlike the temporal mean baseline.
- During the event pride parade, the traffic on 5th Avenue was all diverted to other routes, creating an anomaly in the traffic patterns. Therefore, we saw a dip in the traffic counts. Similar observation as the snow day, the temporal mean baseline does not recover the missing values. However, even though the inpainting results from our model are close to the ground truth values, they slightly overestimate the results.

Overall, the reconstruction accuracy is compelling at specific locations, but not perfect. For 5th Avenue scenario, the parade can be seen as an anomaly, which is rare in the training stage and hard to detect. But this scenario represents another potential application: by interpreting the parade as a data error to correct, we can use the masking to replace known bad data with a semantically consistent alternative. This “airbrushing” of flaws in the data (from, say, faulty sensors) can be used to improve the quality of training sets for downstream applications. Penn Station represents a high-demand area for taxis. Our model tends to underestimate the high demand at this location, though biased masking improves the prediction. For Lower East Side, there are a few anomalous spikes, to which the proposed models are responsive. For airport and Astoria, our models are no better than the temporal mean approach. We conjecture that the variance of airport traffic confounds prediction in general; localized weighting may improve performance. For Astoria, the vanishingly low demand is harder to predict.

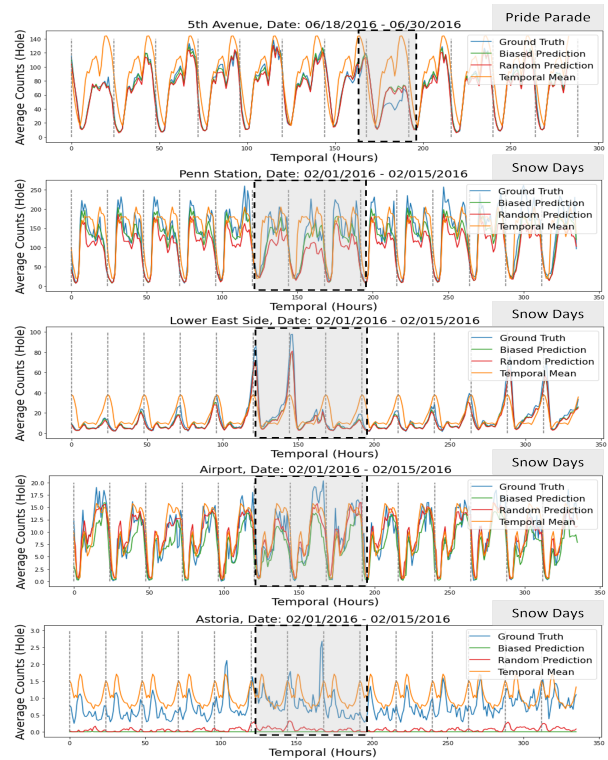


Figure 6: Temporal line plots of evaluations for five scenarios. In each plot, we visualize the ground truth, prediction from model trained with biased masking and random masking, and predictions from temporal mean method. Two evaluation periods, Feb. and June are selected. The irregular events, extreme snow days and pride parade, are annotated with grey regions.

6. Discussion

Our study is motivated by the inconsistent availability of urban data caused by missing, corrupt, or inaccurate data, which hinders their use in downstream tasks, especially learning tasks, that require coverage and accuracy. We designed and implemented a model based on partial convolutions that can tolerate irregular missing regions. Our contributions include: 1) To capture the temporal dependency in urban data, we employed 3D partial convolutional model. Qualitatively, we confirmed from the visual examples that image inpainting techniques can be used to reconstruct data in large, irregular regions in space and time. Quantitatively, we confirmed that extending the model architecture to 3D improves performance. 2) We experimented with varying the number of timesteps per training sample, finding non-trivial tradeoffs — increasing the temporal dimension to a certain threshold improves performance in general, regardless of masking strategy. 3) To address the spatial skew in urban mobility data, we proposed a biased masking approach

which encourages the model to attend to dense, dynamic regions. The model learns faster and is not rewarded for accurate predictions in trivially inactive areas. Biased masking showed improved performance across all values of T , multiple global evaluation strategies, and most local evaluation scenarios. 4) We evaluated performance in local settings, demonstrating that the model is not just learning an average value, but is responsive to subtle spatial variation. The model captures irregular traffic patterns caused by transient events, such as extreme weather and the Pride Parade, and showed that biased masking can improve performance in local settings.

Besides direct imputation, applications may include forecasting, transfer learning between cities, and uncovering relationships between variables. Our focus in this paper is human mobility (taxi rides, bike share trips), the framework is applicable to any dataset of human activity governed by population density.

7. Limitations & Future Work

There are several limitations of our study that represent directions for future work. First, our results on mobility data may extend to other urban activity (e.g., 911 calls, crowd movement, business permits, public safety events, housing events, and more). We do not consider the generalizability of these methods to multiple variables, or variables that do not follow the same spatial patterns; there are opportunities to exploit correlations between variables to improve performance. Additionally, the taxi dataset is exceptionally large and complete; understanding how these techniques behave in low-data regimes is important for practical applications. Integration of masked multi-variate data may be an opportunity: given the shared built environment, models trained on one variable may transfer to predictions of other variables. Second, rasterizing event data to a form amenable to computer vision techniques involves a number of design choices we did not study: resolution, overlap, and irregular boundaries may present opportunities or challenges. In particular, data associated with census blocks, tracts, or individual trajectories lose information when regridded as histograms. In these cases, graph neural networks may be more appropriate to represent the spatial adjacency relationships.

References

[1] Sebastian Abt. “The Missing Data Problem in Cyber Security Research”. In: 8. *GI FG SIDAR Graduierten-Workshop über Reaktive Sicherheit.*: 2013, p. 8.

[2] Bumjoon Bae et al. “Missing data imputation for traffic flow speed using spatio-temporal cokriging”. In: *Transportation Research Part C: Emerging Technologies* 88 (2018), pp. 124–139. ISSN: 0968-090X. DOI: <https://doi.org/10.1016/j.trc.2018.01.015>.

[3] Bert Blocken. “Computational Fluid Dynamics for urban physics: Importance, scales, possibilities, limitations and ten tips and tricks towards accurate and reliable simulations”. In: *Building and Environment* 91 (2015). Fifty Year Anniversary for Building and Environment, pp. 219–245. ISSN: 0360-1323. DOI: <https://doi.org/10.1016/j.buildenv.2015.02.015>.

[4] Rodrigo C. Carlson et al. “Optimal mainstream traffic flow control of large-scale motorway networks”. In: *Transportation Research Part C: Emerging Technologies* 18.2 (2010), pp. 193–212. ISSN: 0968-090X. DOI: <https://doi.org/10.1016/j.trc.2009.05.014>.

[5] Chenyi Chen et al. “The retrieval of intra-day trend and its influence on traffic prediction”. In: *Transportation Research Part C-emerging Technologies* 22 (2012), pp. 103–118. URL: <https://api.semanticscholar.org/CorpusID:110245023>.

[6] NH Choudhury, A Rahman, and S Ferdousi. “Kriging infill of missing data and temporal analysis of rainfall in North Central region of Bangladesh”. In: *J. Climatol. Weather Forecast* (2015).

[7] Sebastian Daberdaku, Erica Tavazzi, and Barbara Di Camillo. “Interpolation and K-Nearest Neighbours Combined Imputation for Longitudinal ICU Laboratory Data”. In: *2019 IEEE International Conference on Healthcare Informatics (ICHI)*. 2019, pp. 1–3. DOI: [10.1109/ICHI.2019.8904624](https://doi.org/10.1109/ICHI.2019.8904624).

[8] Yongshun Gong et al. “A spatial missing value imputation method for multi-view urban statistical data”. In: *Proceedings of the Twenty-Ninth International Conference on International Joint Conferences on Artificial Intelligence*. 2021, pp. 1310–1316.

[9] George Grekousis. “Artificial neural networks and deep learning in urban geography: A systematic review and meta-analysis”. In: *Computers, Environment and Urban Systems* 74 (2019), pp. 244–256.

- [10] Seungwoong Ha and Hawoong Jeong. “Unraveling hidden interactions in complex systems with deep learning”. In: *Scientific reports* 11.1 (2021), pp. 1–13.
- [11] Bill Howe et al. “Integrative urban AI to expand coverage, access, and equity of urban data”. In: *The European physical journal. Special topics* 231.9 (2022), pp. 1741–1752. ISSN: 1951-6355. DOI: 10 . 1140 / epjs / s11734 - 022 - 00475-z.
- [12] Song-Hee Kang, Youngjin Choi, and Jae Young Choi. “Restoration of Missing Patterns on Satellite Infrared Sea Surface Temperature Images Due to Cloud Coverage Using Deep Generative Inpainting Network”. In: *Journal of Marine Science and Engineering* 9.3 (2021), ISSN: 2077-1312. DOI: 10 . 3390 / jmse9030310.
- [13] Guilin Liu et al. *Image Inpainting for Irregular Holes Using Partial Convolutions*. 2018. arXiv: 1804.07723 [cs.CV].
- [14] Kai Liu, Junke Li, and Syed Sabahat Hussain Bukhari. “Overview of Image Inpainting and Forensic Technology”. In: *Security and Communication Networks* 2022 (2022),
- [15] Eric Hsueh-Chan Lu and Chia-Yu Liu. “A Spatial-Temporal Approach for Air Quality Forecast in Urban Areas”. In: *Applied Sciences* 11.11 (2021), ISSN: 2076-3417. DOI: 10 . 3390 / app11114971.
- [16] Xiaolei Ma et al. *Learning Traffic as Images: A Deep Convolutional Neural Network for Large-Scale Transportation Network Speed Prediction*. 2017. arXiv: 1701 . 04245 [cs.LG].
- [17] Luis Moreira-Matias et al. “Predicting taxi-passenger demand using streaming data”. In: *IEEE Transactions on Intelligent Transportation Systems* 14.3 (2013), pp. 1393–1402.
- [18] Deepak Pathak et al. *Context Encoders: Feature Learning by Inpainting*. 2016. arXiv: 1604 . 07379 [cs.CV].
- [19] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. “U-Net: Convolutional Networks for Biomedical Image Segmentation”. In: *CoRR* abs/1505.04597 (2015), arXiv: 1505 . 04597. URL: <http://arxiv.org/abs/1505.04597>.
- [20] Bilong Shen et al. “StepDeep: A Novel Spatial-Temporal Mobility Event Prediction Framework Based on Deep Neural Network”. In: *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining. KDD '18*. London, United Kingdom: Association for Computing Machinery, 2018, pp. 724–733. ISBN: 9781450355520. DOI: 10 . 1145 / 3219819 . 3219931.
- [21] Felipe de Souza, Omer Verbas, and Joshua Auld. “Mesoscopic Traffic Flow Model for Agent-Based Simulation”. In: *Procedia Computer Science* 151 (2019). The 10th International Conference on Ambient Systems, Networks and Technologies (ANT 2019) / The 2nd International Conference on Emerging Data and Industry 4.0 (EDI40 2019) / Affiliated Workshops, pp. 858–863. ISSN: 1877-0509. DOI: <https://doi.org/10.1016/j.procs.2019.04.118>.
- [22] Roman Suvorov et al. “Resolution-robust large mask inpainting with fourier convolutions”. In: *Proceedings of the IEEE/CVF winter conference on applications of computer vision*. 2022, pp. 2149–2159.
- [23] Jarin Tasnim and Debajyoti Mondal. “Data Reduction and Deep-Learning Based Recovery for Geospatial Visualization and Satellite Imagery”. In: *2020 IEEE International Conference on Big Data (Big Data): 2020*, pp. 5276–5285. DOI: 10 . 1109 / BigData50022.2020.9378008.
- [24] Rui Xu et al. “Texture memory-augmented deep patch-based image inpainting”. In: *IEEE Transactions on Image Processing* 30 (2021), pp. 9112–9124.
- [25] Jiahui Yu et al. *Generative Image Inpainting with Contextual Attention*. 2018. arXiv: 1801 . 07892 [cs.CV].
- [26] Qiang Zhang et al. “Missing Data Reconstruction in Remote Sensing Image With a Unified Spatial–Temporal–Spectral Deep Convolutional Neural Network”. In: *IEEE Transactions on Geoscience and Remote Sensing* 56.8 (2018), pp. 4274–4288. DOI: 10 . 1109 / TGRS . 2018 . 2810208.
- [27] Tong Zhou et al. “Learning oracle attention for high-fidelity face completion”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.: 2020, pp. 7680–7689.